

Pham Khanh Quan, Ph.D. Student. Chungbuk National University, Korea

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🌐 <https://phamkhanhquan197.github.io/>

Education

- Sep 2023 – February 2027 📖 **Ph.D. Chungbuk National University** in Information & Communication Engineering. [GPA: 4.41/4.5]
- Sep 2021 – August 2023 📖 **M.Sc. Chungbuk National University** in Information & Communication Engineering. [GPA: 4.3/4.5]
Thesis title: *Experimental Evaluation and Analysis of Federated Learning in Edge Computing Environments*.
- 2015 – 2020 📖 **B.S. Ton Duc Thang University** in Electronics and Telecommunication Engineering. [GPA: 7.06/10.0]
Thesis title: *On the Implementation of a Low-Cost Mind-Voice-and-Gesture-Controlled Humanoid Robotic Arm Using Leap Motion and Neurosky Sensor*.

Employment History

- July 2020 – July 2021 📖 **AI-Robotic R&D Engineer** in [ARAR](#) (Advanced Robotics & Automation Research)
• Robotics R&D lab specializing in SLAM navigation and automation for service robots.
- Dec 2019 – June 2020 📖 **Engineering Intern** at [Intel Product Viet Nam](#).
• Internship in QA auditing and preventive maintenance compliance.
- Feb 2019 – August 2019 📖 **Engineering Intern** at [Renesas Design Viet Nam](#).
• Semiconductor R&D center focused on embedded software development for chip design.

Research Interests

My research focuses on efficient, adaptive, and deployable AI systems, with an emphasis on large language models (LLMs), parameter-efficient model adaptation, and intelligent decision-making. I have experience developing lightweight adaptation methods, including LoRA and SVD-based low-rank techniques, for efficiently adapting AI models to heterogeneous data and resource-constrained environments. My research also explores personalized and adaptive AI through reinforcement learning (RL), enabling automated optimization and decision-making in dynamic environments. More broadly, my interests include LLM-based intelligent applications, retrieval-augmented generation (RAG), AI system optimization, and the practical deployment of AI solutions for real-world applications.

Research Publications

Journal Articles

- 1 **P. K. Quan** and Co-authors, "Personalized federated learning with spectral allocation and adaptive rank control," *Expert Systems with Applications*, 2026, Under review.
- 2 **P. K. Quan**, N. Khoa, L. Y. Zhang, M. Liyanage, and T. Kim, "Fedkls: KI-guided spectral allocation for large-model federated learning in heterogeneous iot environments," *IEEE Internet of Things Journal*, 2026, Under review.
- 3 **P. K. Quan**, M. Kundroo, G. Ban, S. Bae, and T. Kim, "Reinforcement learning-based laser cutting machine parameter optimization," *Journal of Intelligent Manufacturing*, pp. 1–16, 2025. [DOI](#): [10.1007/s10845-025-02619-z](https://doi.org/10.1007/s10845-025-02619-z)

- 4 V. Van Truong, **P. K. Quan**, D.-H. Park, and T. Kim, "Performance evaluation of decentralized federated learning: Impact of fully and k-connected topologies, heterogeneous computing resources, and communication bandwidth," *IEEE Access*, 2025. [DOI: 10.1109/ACCESS.2025.3542772](#)
- 5 **P. K. Quan** and T. Kim, "Elastic federated learning with kubernetes vertical pod autoscaler for edge computing," *Future Generation Computer Systems*, vol. 158, pp. 501–515, 2024. [DOI: 10.1016/j.future.2024.04.047](#)
- 6 **P. K. Quan**, M. Kundroo, and T. Kim, "Experimental evaluation and analysis of federated learning in edge computing environments," *IEEE Access*, vol. 11, pp. 33 628–33 639, 2023. [DOI: 10.1109/ACCESS.2023.3262945](#)
- 7 **P. K. Quan**, T. V. Vo, and P. T. Tran, "On the implementation of a low-cost mind-voice-and-gesture-controlled humanoid robotic arm using leap motion and neurosky sensor," *Journal of Electrical Engineering & Technology*, vol. 17, no. 1, pp. 665–683, 2022. [DOI: 10.1007/s42835-021-00903-5](#)

Conference Proceedings

- 1 V. V. Truong, **P. K. Quan**, and T. Kim., "Flexified: Adaptive resource-based client topology selection and similarity-based aggregation in decentralized federated learning," in *Conference of Korean Artificial Intelligence Association (CKAIA)*, 2024.
- 2 **P. K. Quan** and T. Kim., "The impact of data-rich clients in non-iid federated learning," in *Korea Digital Contents Society Summer Academic Conference (DCS)*, 2023.

Skills

Languages	English, Korean, Vietnamese
Coding	Python, C/C++, LaTeX
Web Dev	HTML

Miscellaneous Experience

Awards and Achievements

- 2025 **Outstanding Graduate Research Award**, Chungbuk National University.
- 2020 **Second Place Winner at Student Research of Science**, Ton Duc Thang University.

References

Prof. Taehong Kim

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Prof. Tran Thanh Phuong

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